Uncovering Data Landscapes through Data Reconnaissance and Task Wrangling

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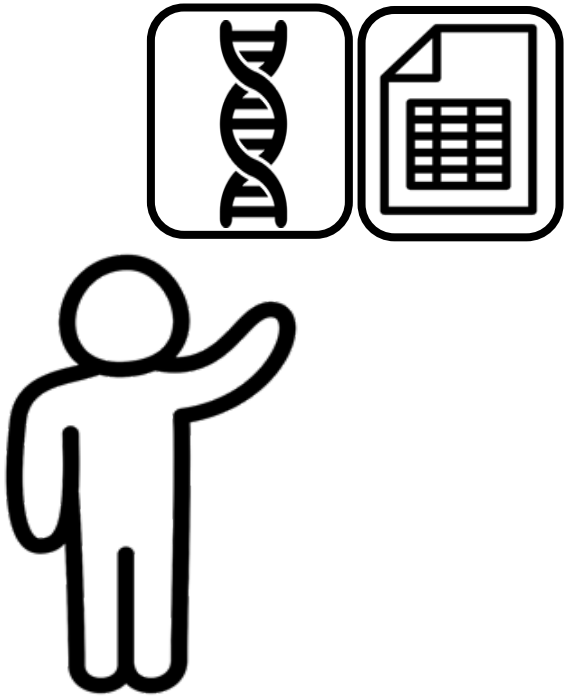
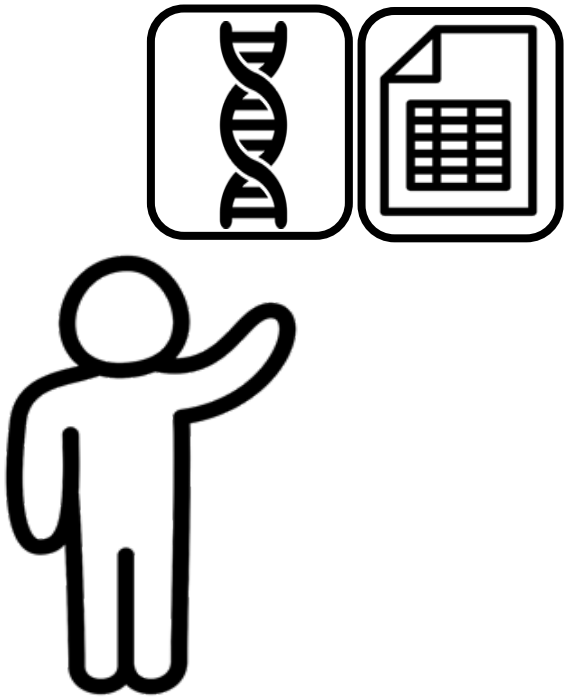
Domain Expert’s Data



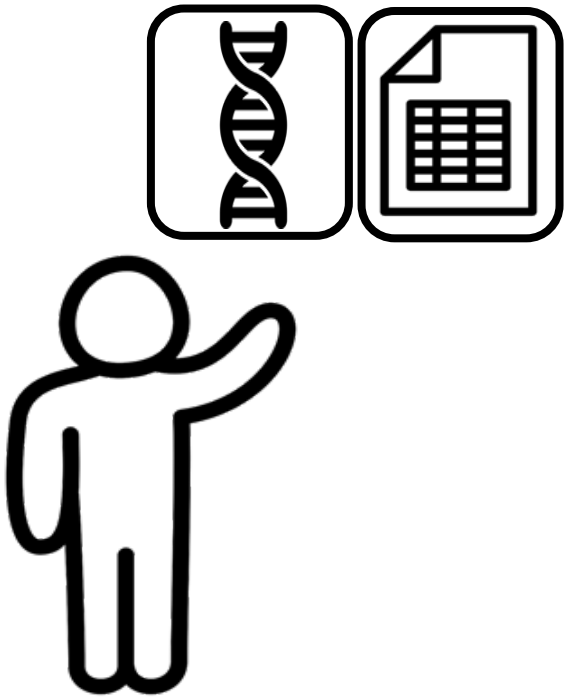


Unexplored data

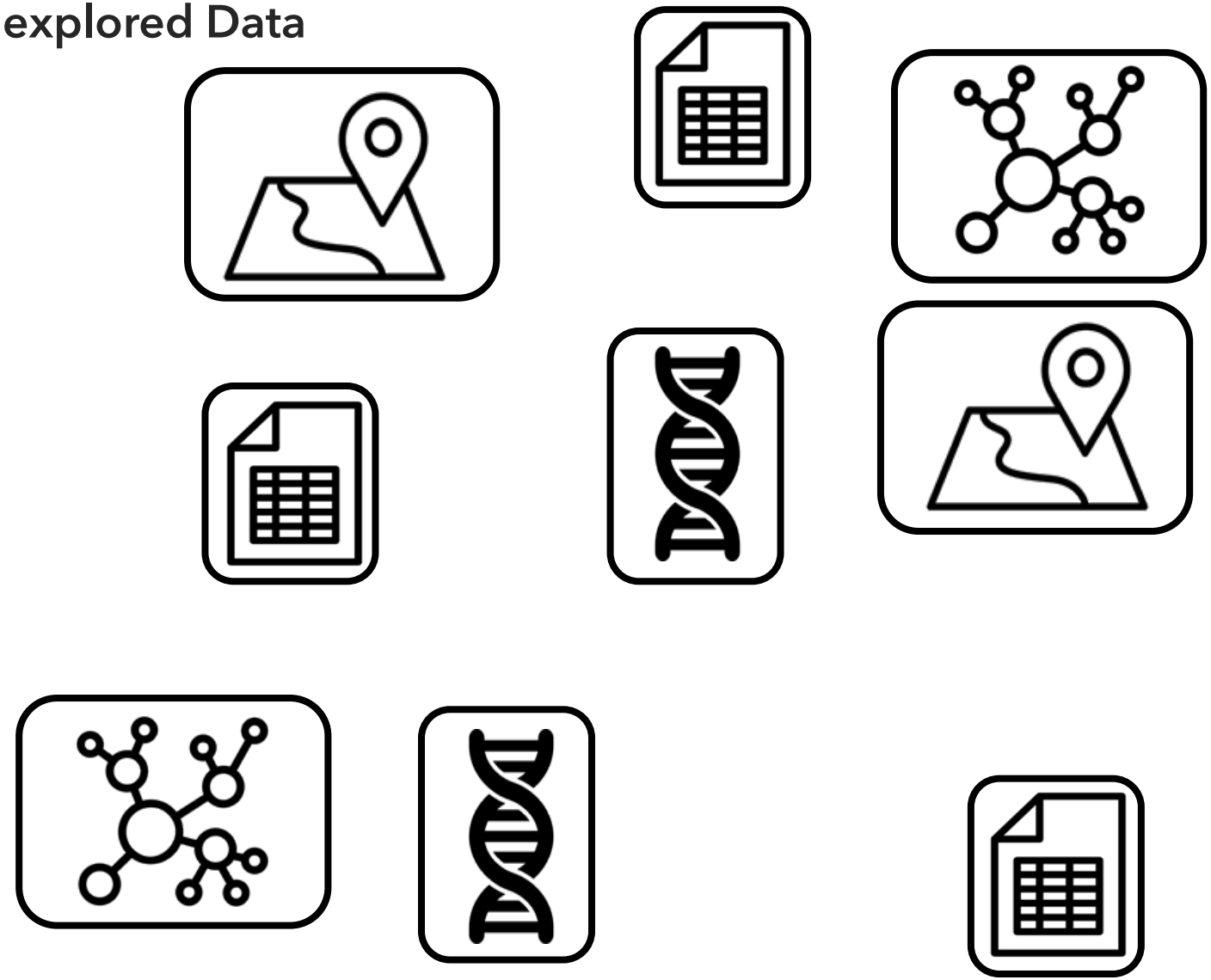
Data domain expert <u>does not have</u>
but could be useful for analysis

Data landscape : collection of heterogeneous datasets

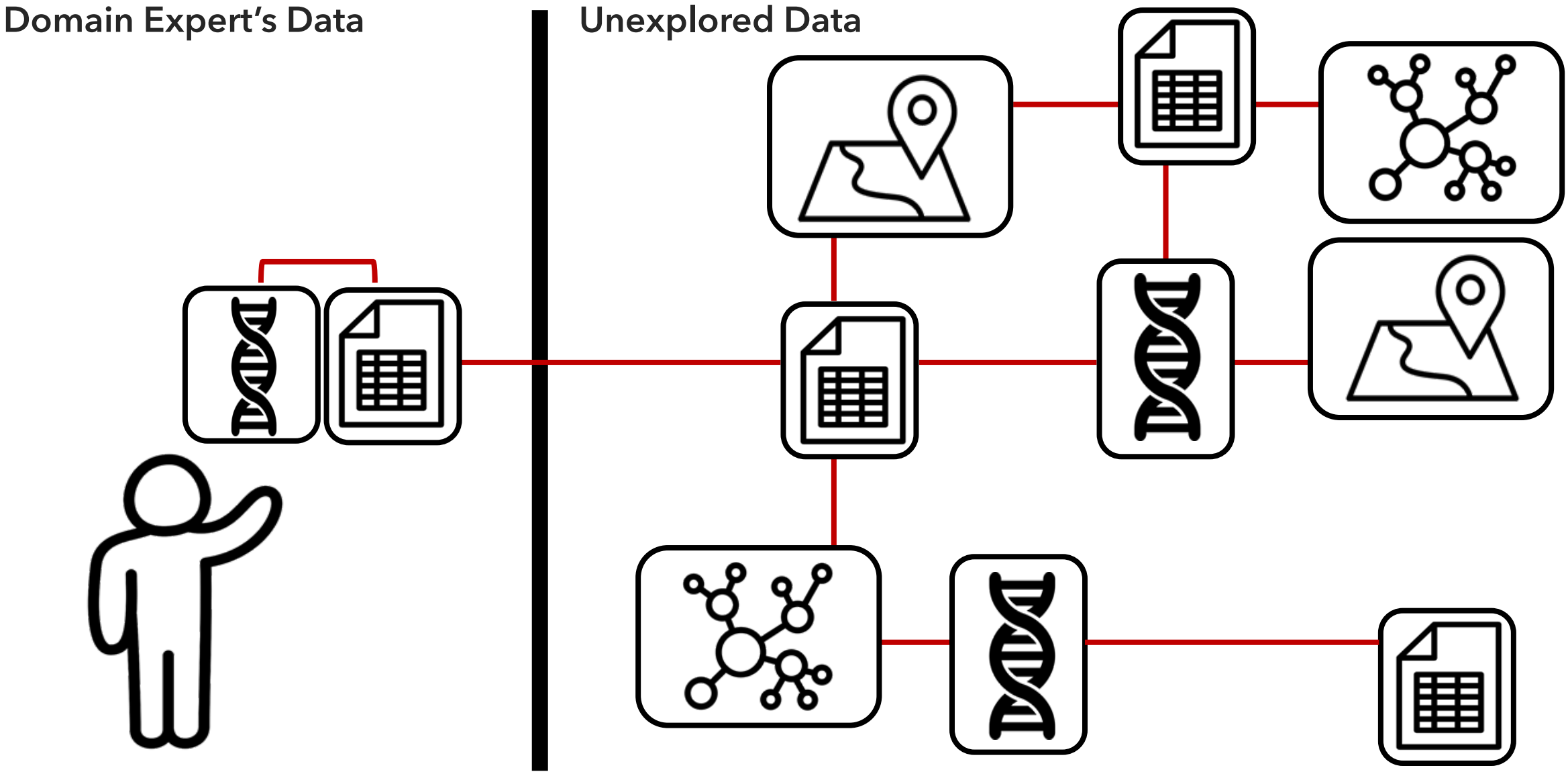
Domain Expert's Data



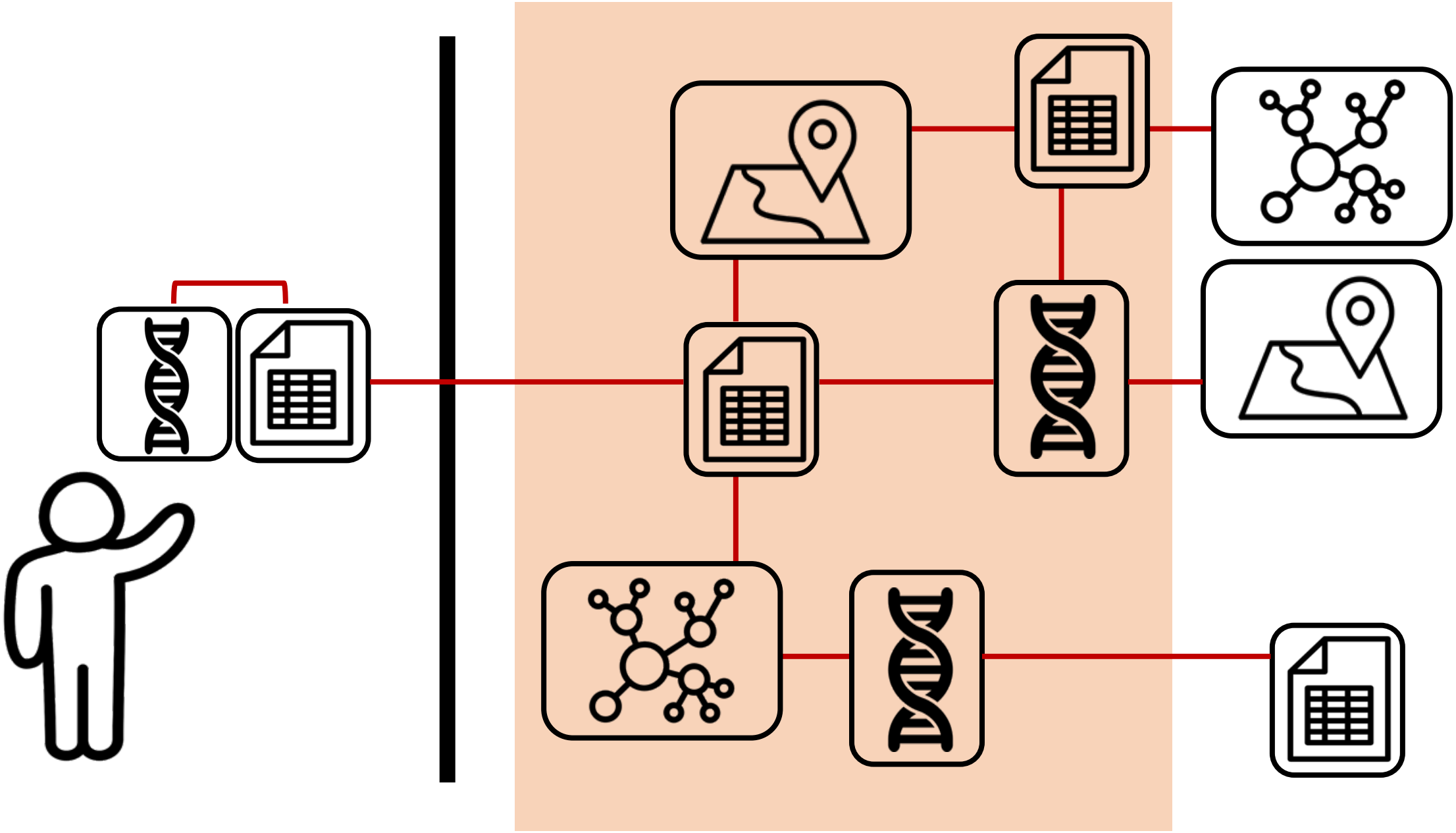
Unexplored Data



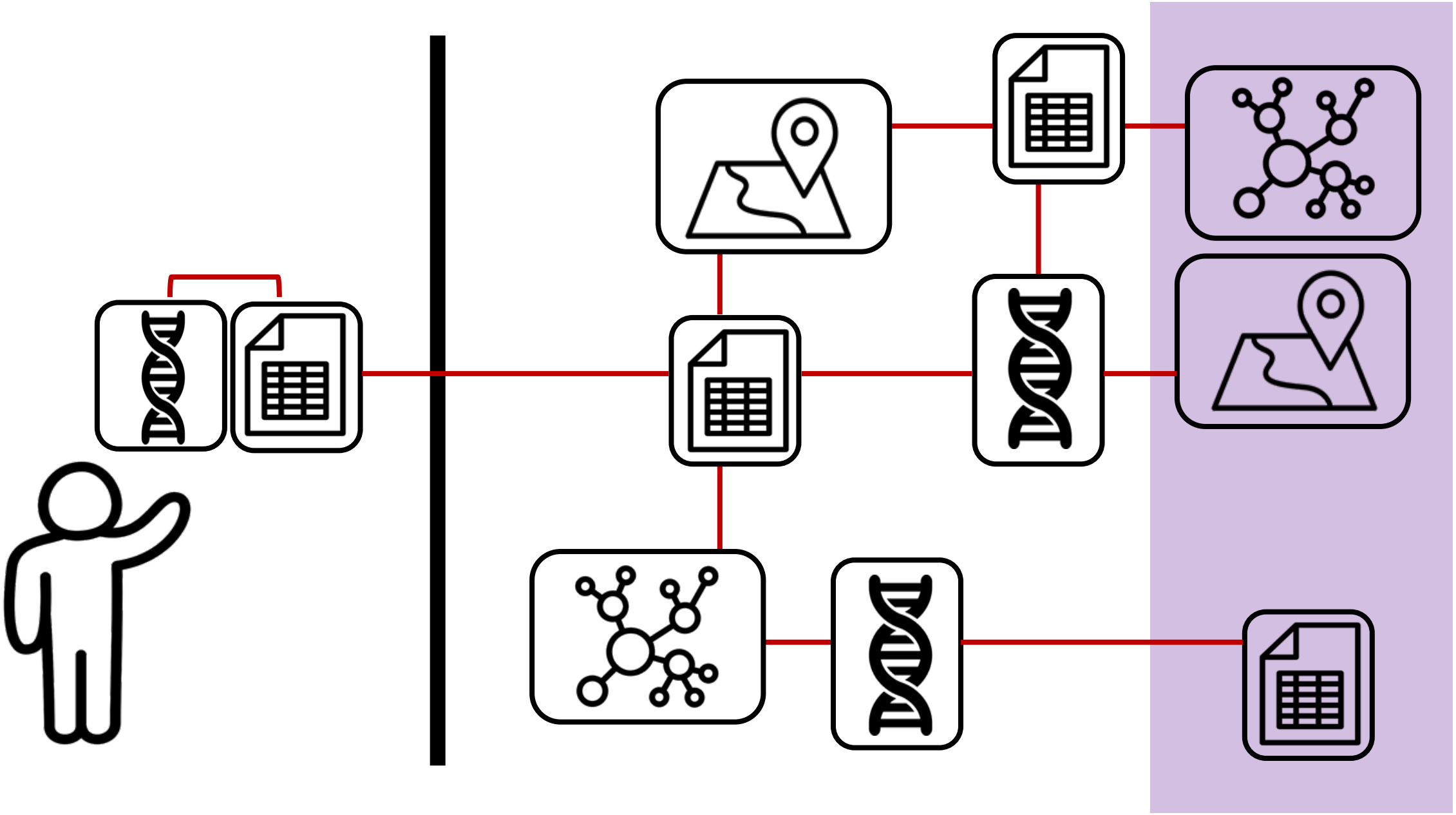
Datasets can be connected through shared attributes



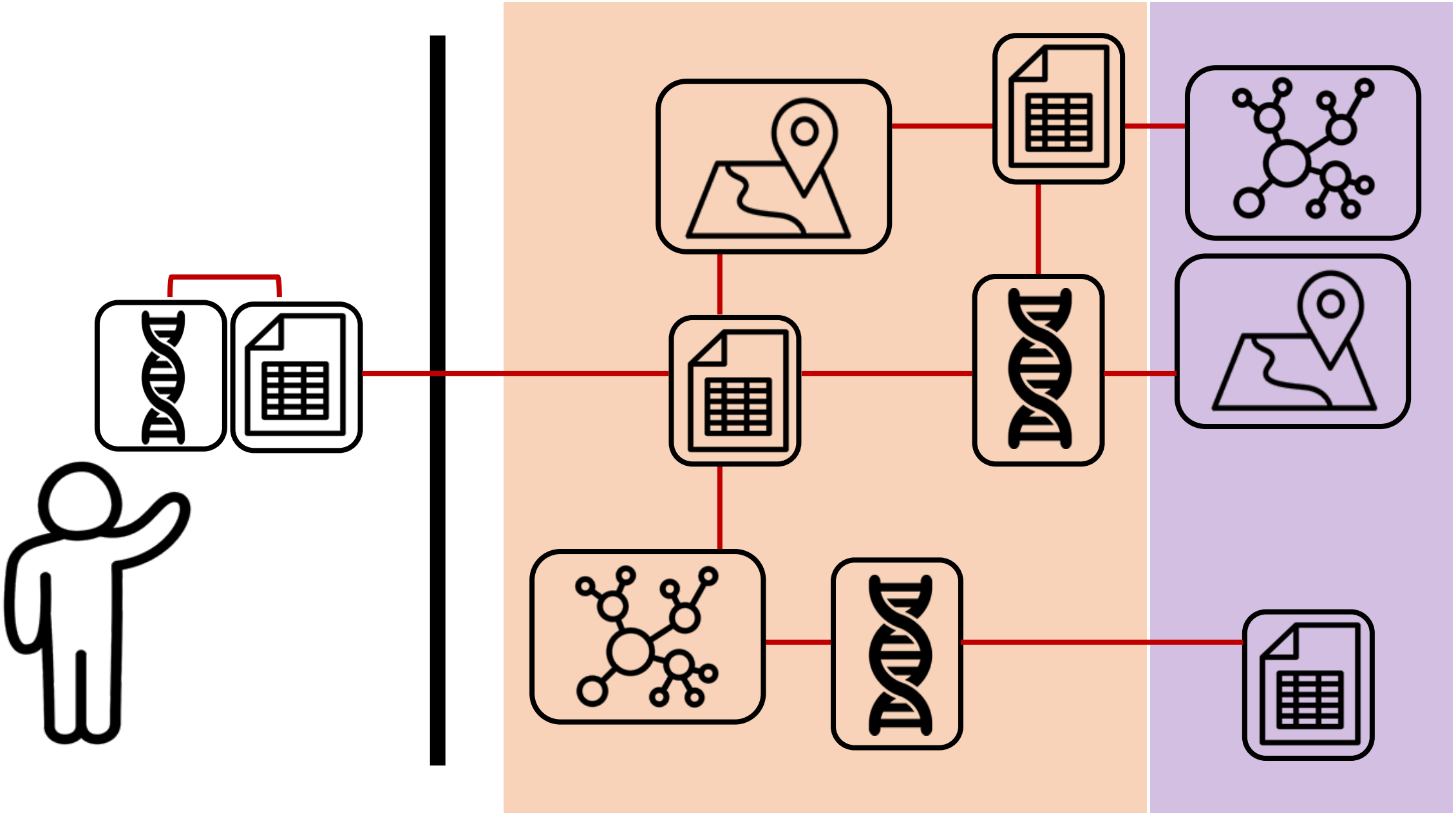
Experts may not have access to all of data

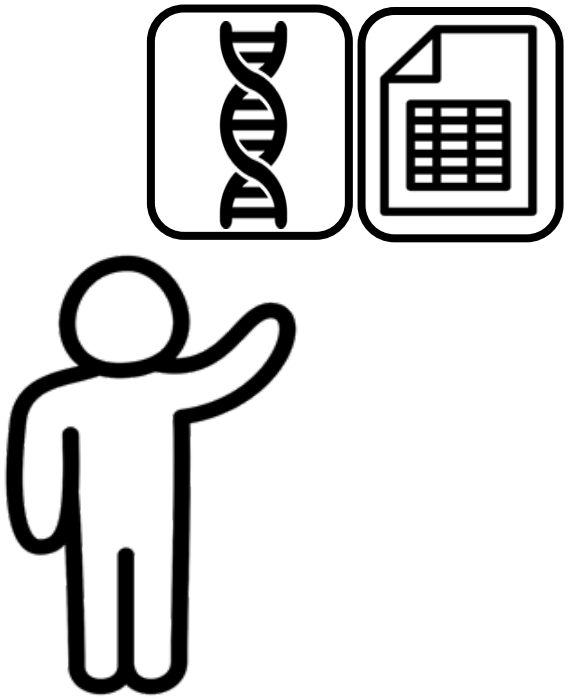


Experts may have not yet uncovered some data



Data landscapes are complex and overwhelming





Domain experts need help uncovering and reasoning about heterogeneous data landscapes

Contribution :

Operational definitions for data reconnaissance and task wrangling

Two interrelated processes uncover data landscapes:

Data Reconnaissance

the process of uncovering an unfamiliar data landscape, including known, unknown, available and unavailable dataset

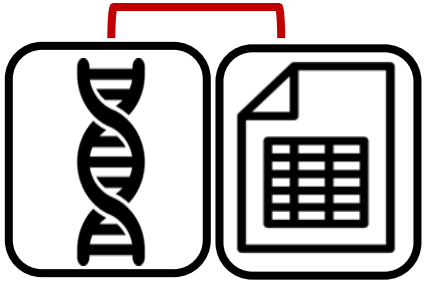
Task Wrangling

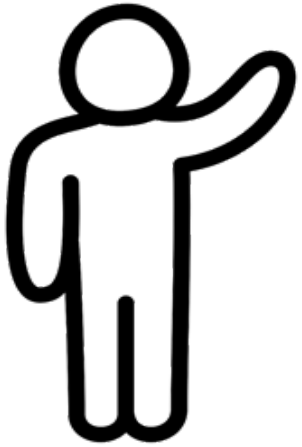
the process of progressively forming a crisper notion of tasks and assessing whether available and known datasets are suitable

Two interrelated processes uncover data landscapes:

Data Reconnaissance

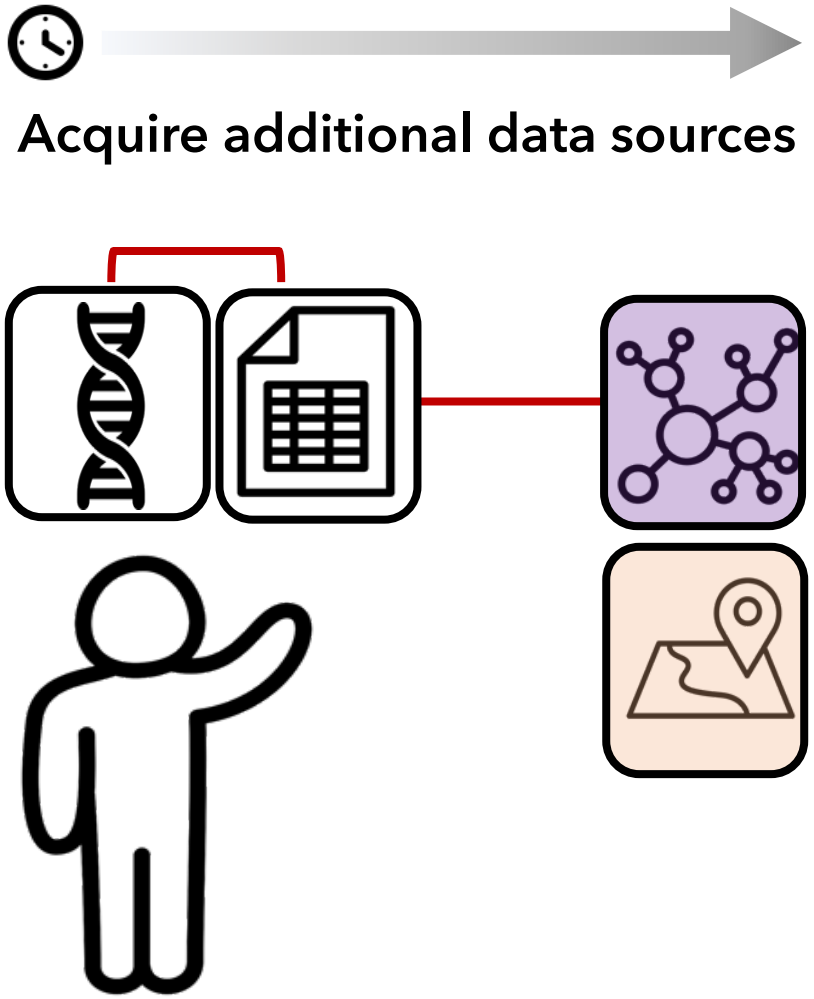
Some Data





Two interrelated processes uncover data landscapes:

Data Reconnaissance



Analysis & visualizations of available data sources supports acquisition of new data:

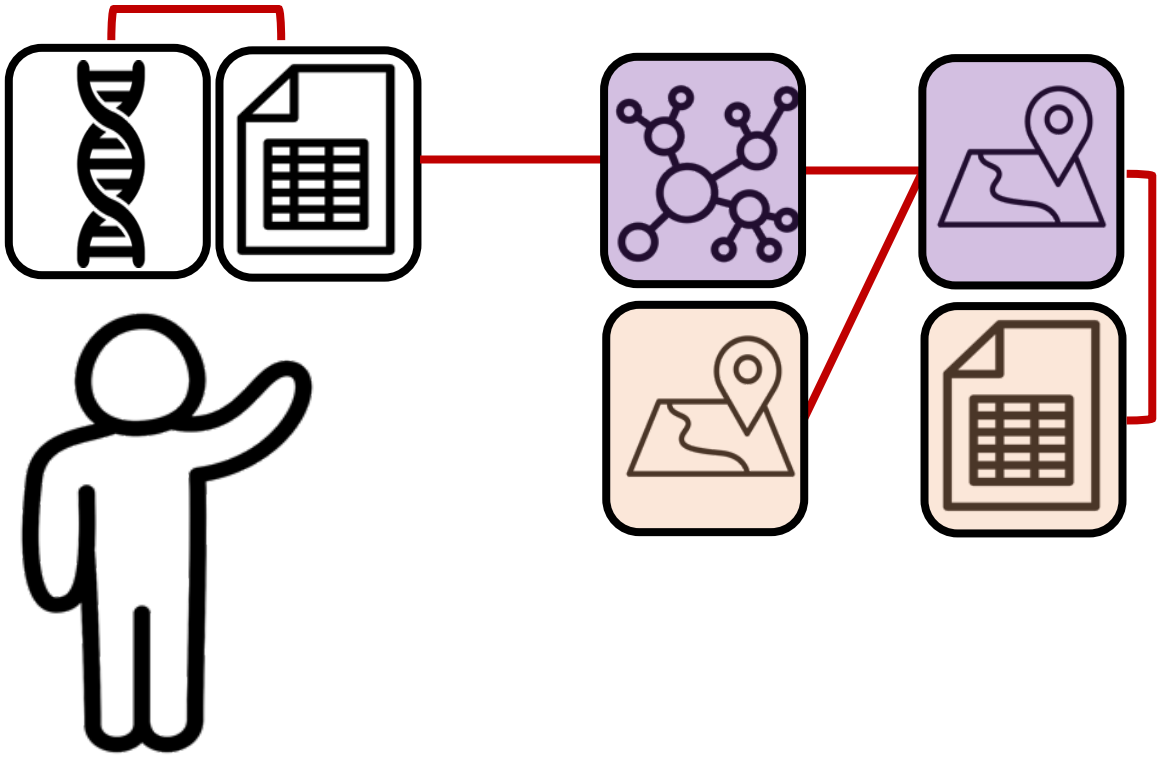
Acquire new, formerly hidden, dataset

Acquire available, but restricted, dataset

Two interrelated processes uncover data landscapes:

Data Reconnaissance

Arrive at a finalized data set



Finalized dataset can be
Analyzed & visualized in depth

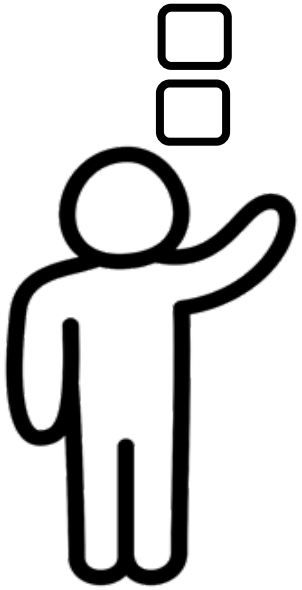
Two interrelated processes uncover data landscapes:

Data Reconnaissance

Task Wrangling

Low Task Clarity

" I don't really know what this data is"



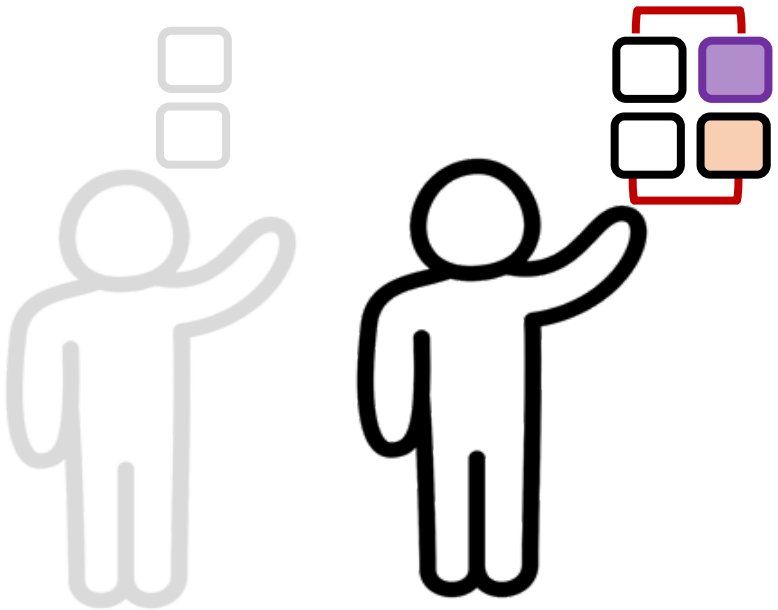
Two interrelated processes uncover data landscapes:

Data Reconnaissance

Task Wrangling

Evolving Task Clarity

" I think might want to see geographic patterns"



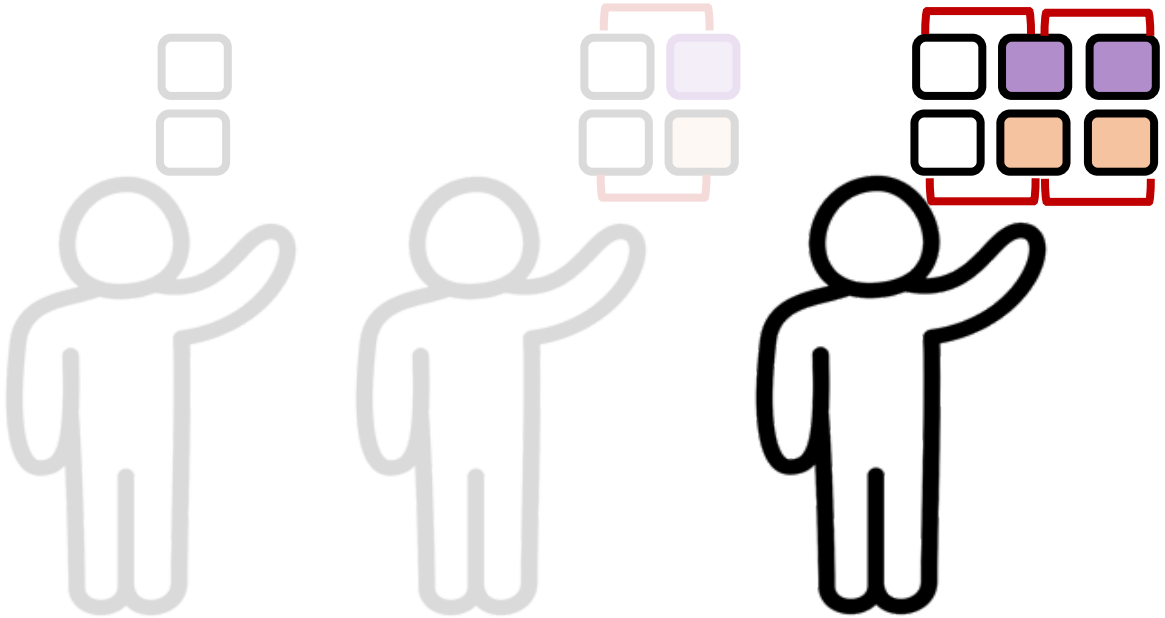
Two interrelated processes uncover data landscapes:

Data Reconnaissance

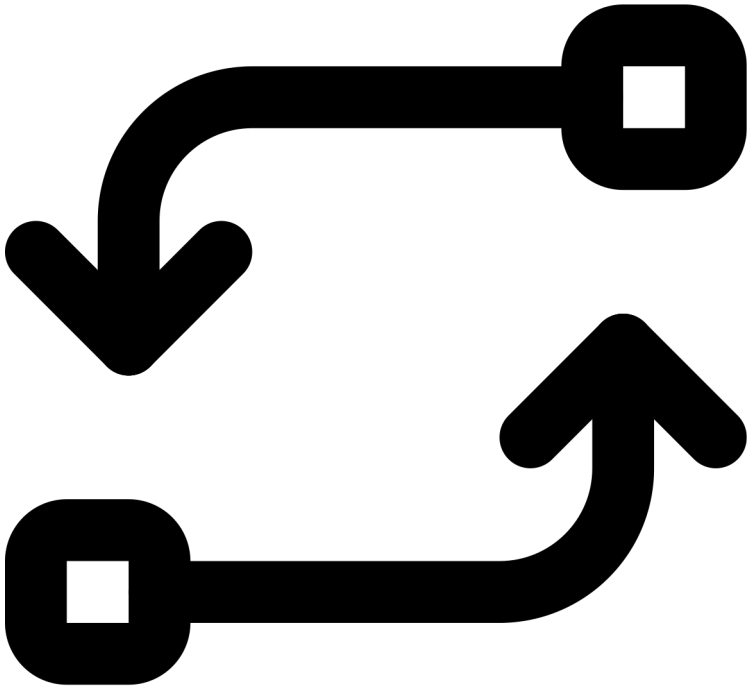
Task Wrangling

Refined Task Clarity

"I want to see the geographic relatedness of connected genomic clusters over time"



Processes influence each other over time



Data Reconnaissance

Data access inspires & refines tasks

Task Wrangling

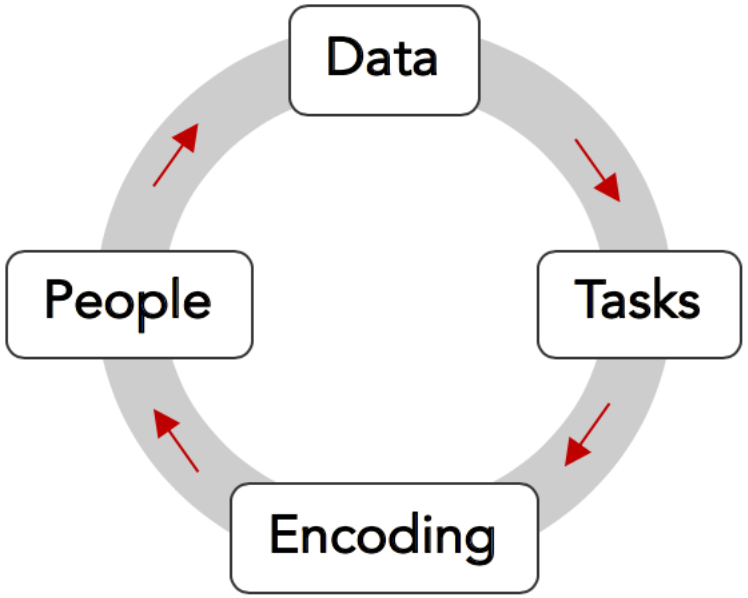
Refined tasks guide the pursuit of data

Contribution :

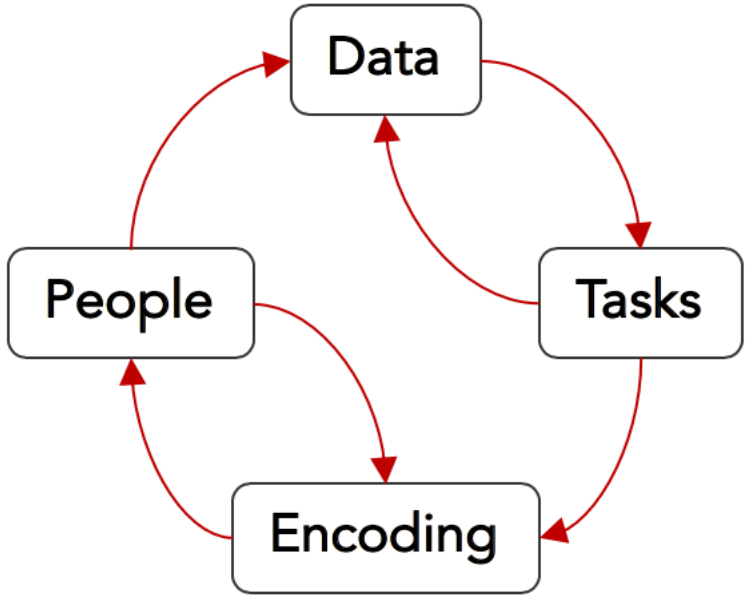
A conceptual framework for
data reconnaissance and task wrangling

Existing methods require crisper tasks & data

Human Centered Design



Design Study Methodology



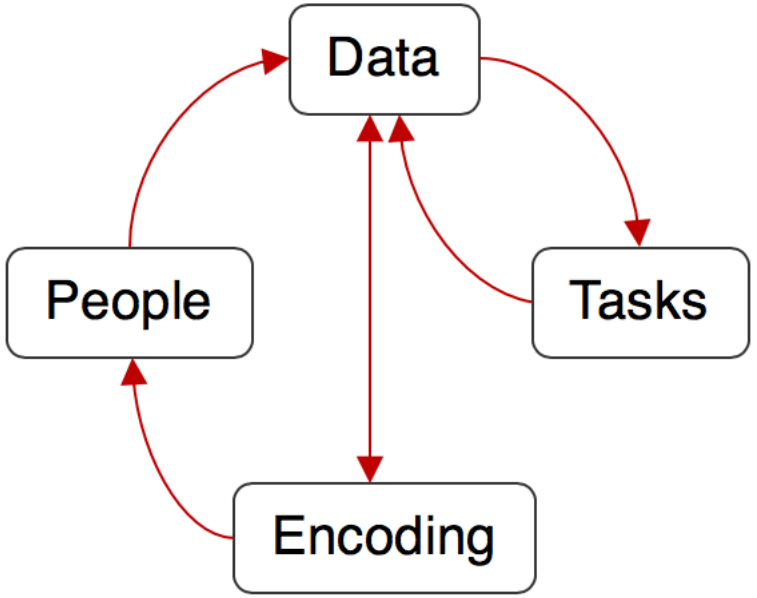
Infovis design methods can be slow in the face of changing data and tasks

Our framework : visualization to uncover data & tasks

Human Centered Design

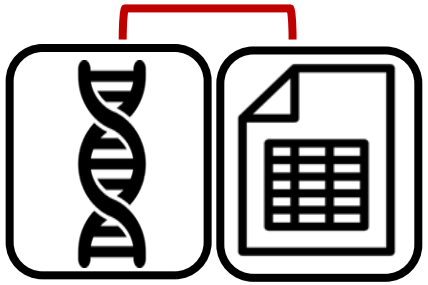
Design Study Methodology

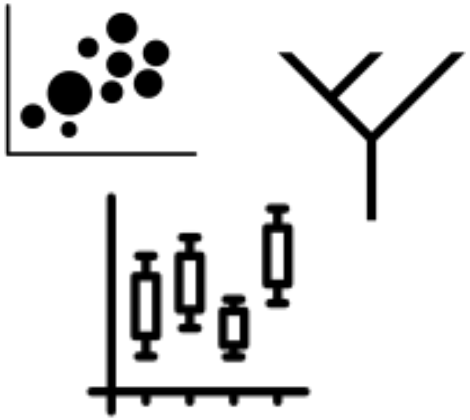
Our framework

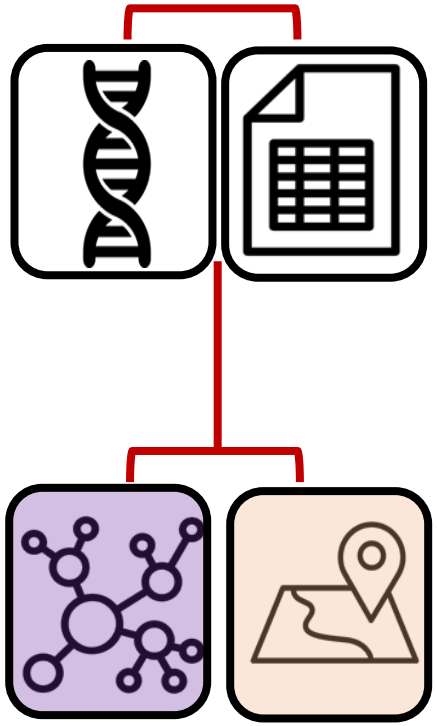


Steps in our conceptual framework

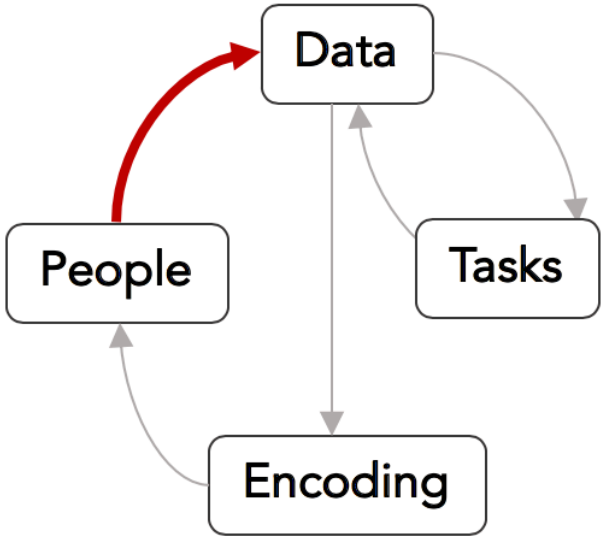
Acquire → View → Assess → Pursue

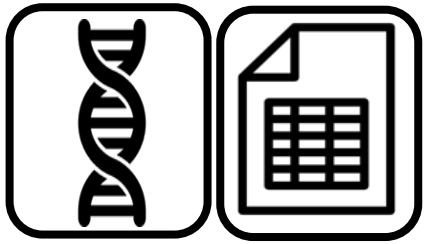




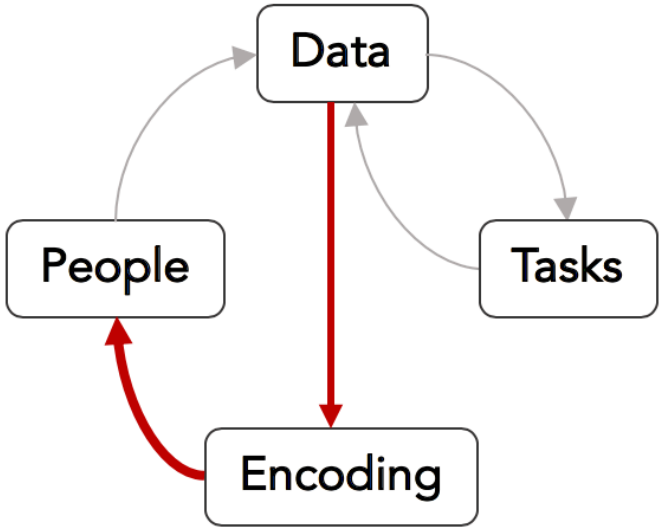


Acquire

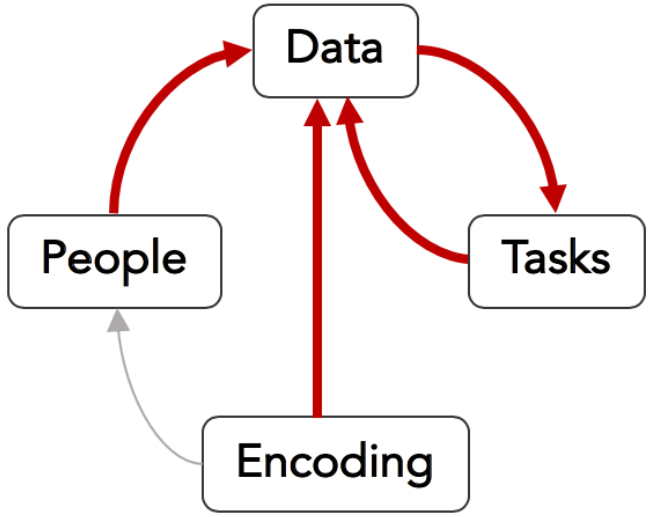




Acquire → View



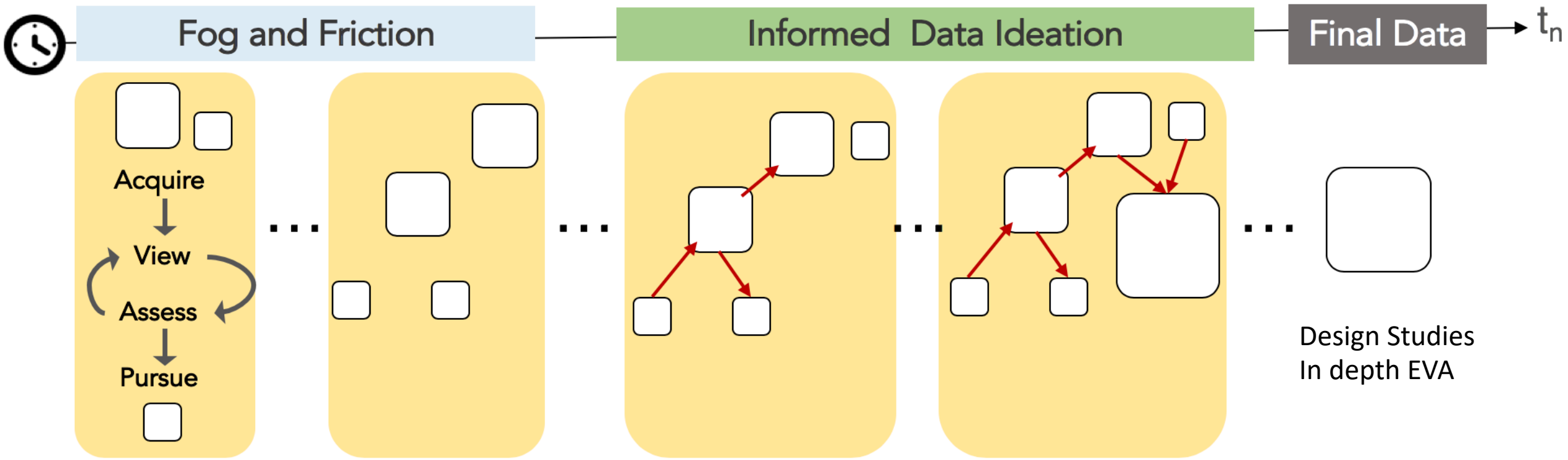
Acquire → View → Assess



Pursue

Acquire → View → Assess

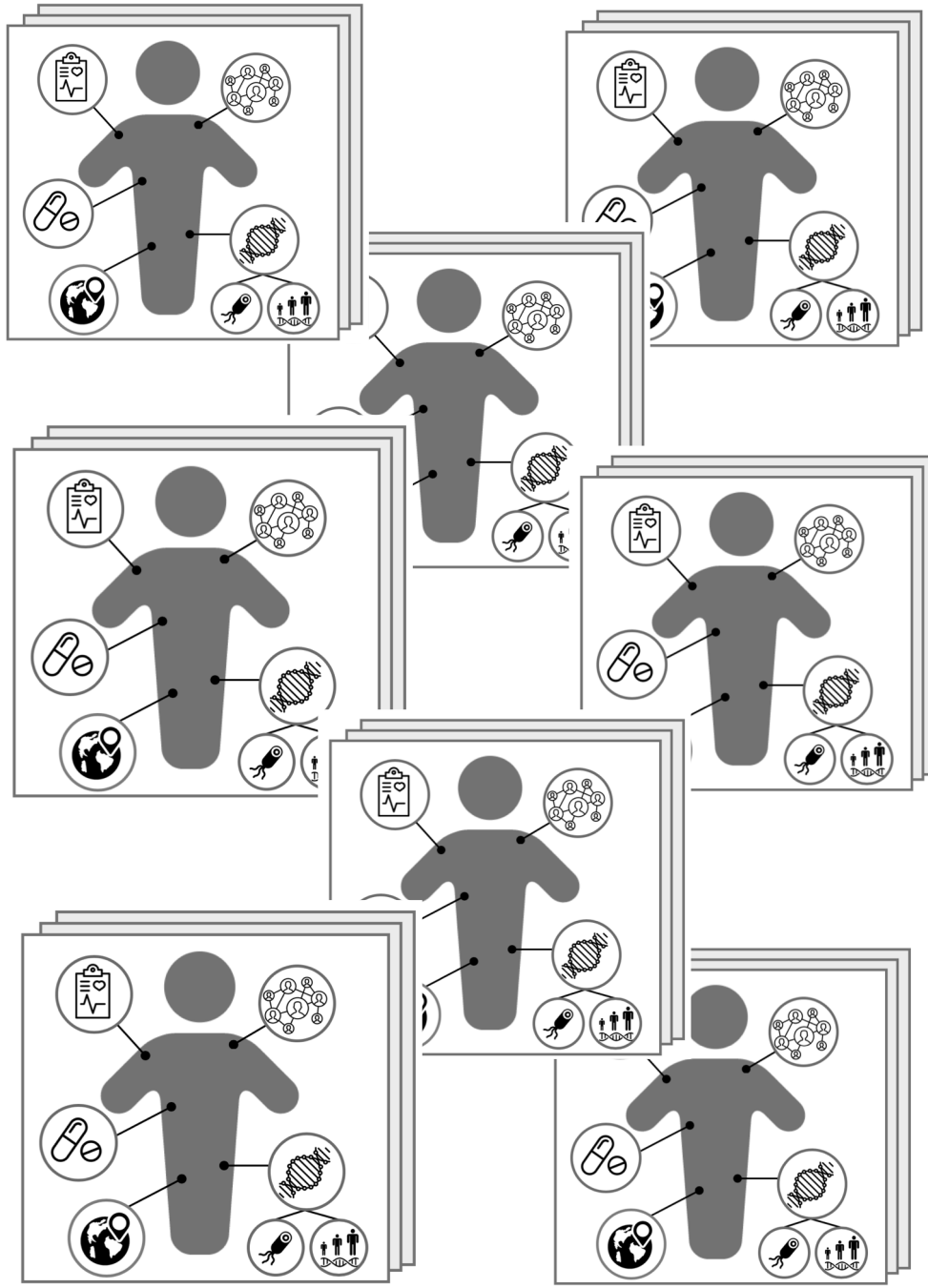
From unknown landscape to the final dataset



Contribution :

Retrospective analysis of design studies to identify data reconnaissance and task wrangling

Observed these challenges first hand with domain experts in genomic epidemiology



“multimodal social networks are not a well established concept even in social science”

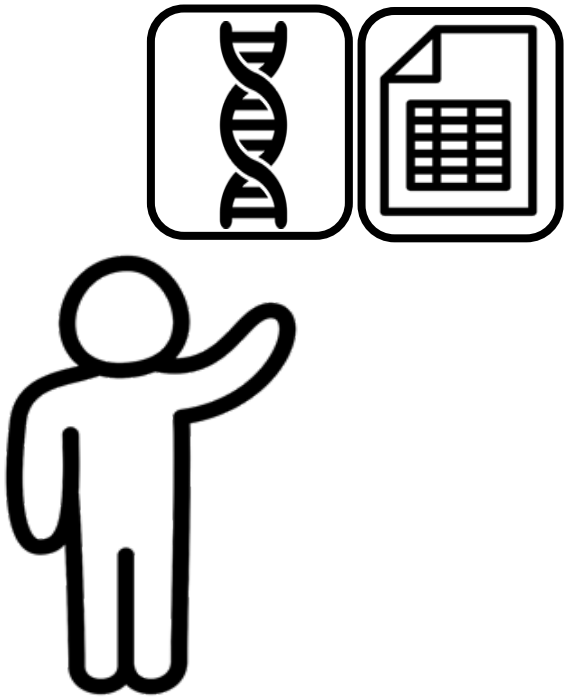
Ghani et al. 2013

Visual Analytics for Multimodal Social Network Analysis: A Design Study with Social Scientists

Have prior design studies also stumbled across data reconnaissance and task wrangling?

Do prior design studies show signs of DR & TW

- Reviewed 100 design studies
- Extracted 235 unique tasks connected to all data reported
- Classified data and tasks to identify DR & TW
- At least 20% of tasks indicative of DR & TW
- Limitations to this approach, but interesting beginning



How can infovis research help domain experts uncover and reasoning about heterogeneous data landscapes?

How can we help?

Building systems suitable for
data reconnaissance and task wrangling

Requirements for data reconnaissance & task wrangling systems:

- **Vis systems can help data reconnaissance and task wrangling**
- **But these systems need to be :**
  - ***Responsive*** to different data types and ill defined tasks
  - ***Adaptable*** to the datavis needs arising different data types
  - ***Relevant*** to domain-specific needs
  - ***Helpful*** to navigate data landscape

THANKS:

Dr. Jennifer Gardy

+ UBC infoVis group

Kimberly Dextras-Romagnino, Madison Elliott, Shannah Fisher, Micheal Opperman, and Zipeng Liu

+ Reviewers and many others that have provided feedback on this work

Pre-Print + Other Stuff

<put this in>

https://amcrisan.github.io/talks

@amcrisan

Workshop plug:

biovis

Wed. Oct 23rd

Posters 67 - 72

Wed. Oct 24th (9:00am -12:20om)

Room 1 : Expert panels, challenge talks